

Summary Report

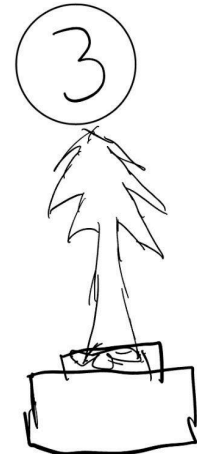
Design Summary

Problem: Some people have lots of trouble thinking of something as simple as a gift with Christmas coming up. This project serves as an easy solution as it can be given to others as a Christmas present and it's not too much effort to think about.

List of working Criteria:

- Led Lights
- Piazzo Speaker music
- 360 motor
- Code
- Mounted Tree
- Push button activation

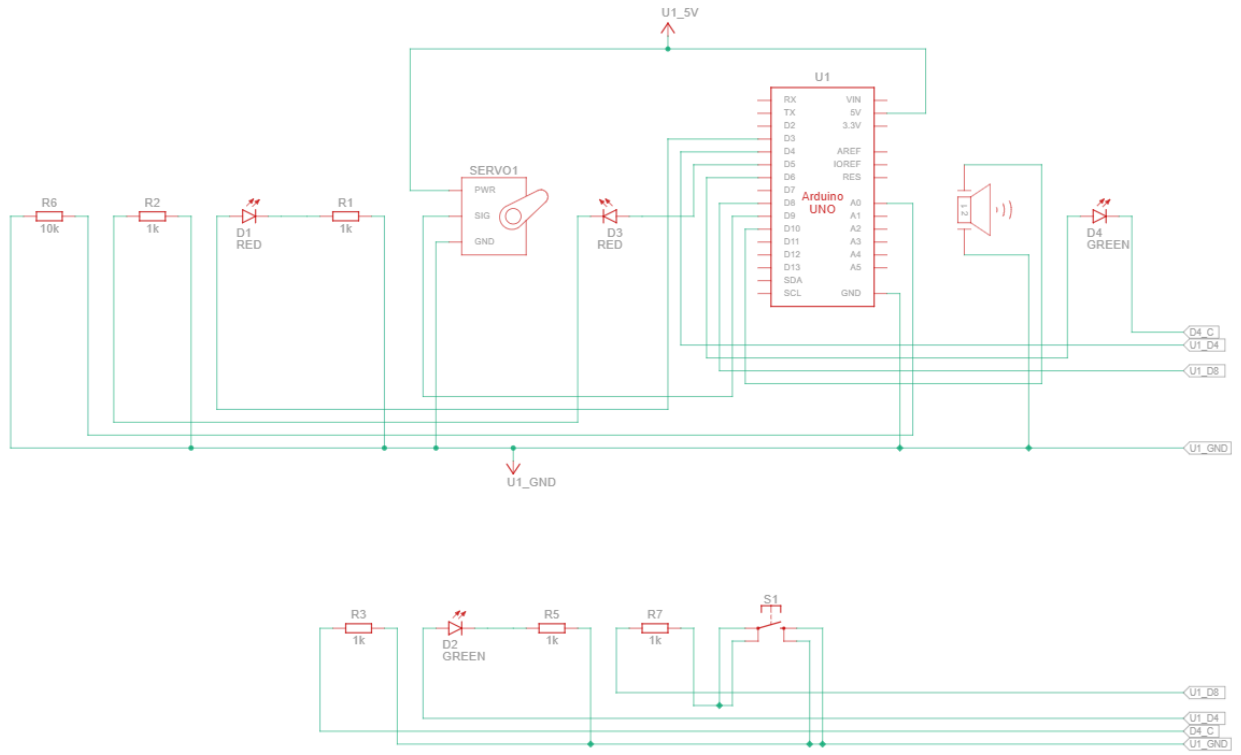
Design Options:



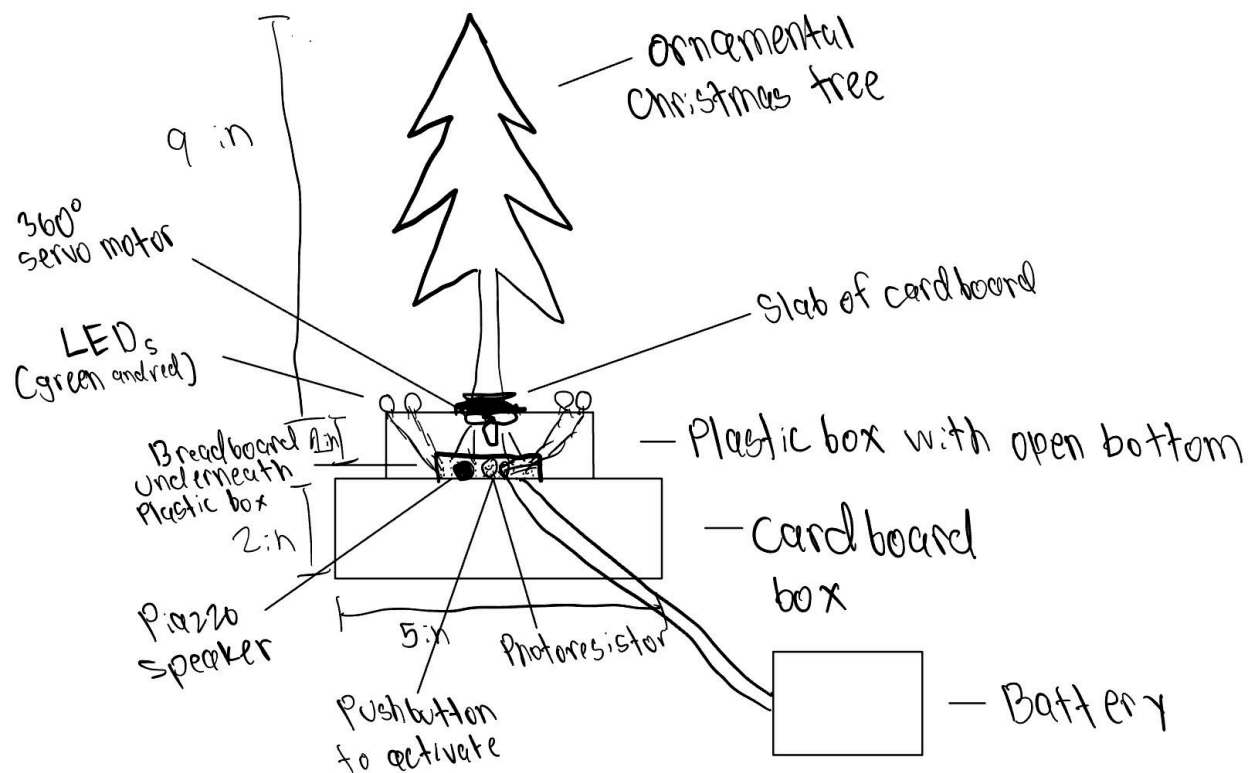
1. Breadboard outside of the main design and wired to connect to the motor inside a box beneath the tree
2. Breadboard UNDER the box that the tree lays upon
3. Breadboard on top of the box

Design Decision: While all three of these designs follow the same principle of the breadboard handling everything related to the tree and motor, the location of the breadboard was very important. We ended up choosing design two for two reasons. First is that it simply looks better when we are unable to see the breadboard that functions the entire design, so we put it inside. Second, wiring would be an extreme issue if we put the breadboard anywhere else other than inside as we would have to find ways to wire all the circuitry from an external source.

Electronic Schematic



Final Design Sketch



Bill Of Materials

[illegible]

Code

```
//library
#include <Servo.h>
//control of the servo motor
Servo myservo;

//pins and what they do
const int servoPin = 9;
const int buzzerPin = 10;
const int buttonPin = 8;
const int ledPins[] = {3, 4, 5, 6};
const int ldrPin = A0;

// the behavior of the photo resistor and
const int STOP_SPEED = 90;
const int SLOW_SPEED = 94;
const int darknessThreshold = 80;

// Music and Melody
const int melody[] = {
  659, 659, 659,
  659, 659, 659,
  659, 784, 523, 587, 659,
  698, 698, 698, 698,
  659, 659, 659,
  587, 587, 659,
  587, 784
};
//duration of the song
const int durations[] = {
  300, 300, 600,
  300, 300, 600,
  300, 300, 400, 400, 700,
  300, 300, 300, 300,
  300, 300, 300,
  600, 700
};
//
const int totalNotes = sizeof(melody) / sizeof(melody[0]);
```

```
38 //
39 const int totalNotes = sizeof(melody) / sizeof(melody[0]);
40
41 // State
42 int currentNote = 0;
43 unsigned long noteStartTime = 0;
44 bool isPlaying = false;
45 bool buttonPreviouslyPressed = false;
46 bool ledsOn = false;
47
48 void setup() {
49   pinMode(buttonPin, INPUT_PULLUP);
50   pinMode(buzzerPin, OUTPUT);
51   for (int i = 0; i < 4; i++) pinMode(ledPins[i], OUTPUT);
52
53   myservo.attach(servoPin);
54   myservo.write(STOP_SPEED);
55   noTone(buzzerPin);
56 }
57
58 void loop() {
59   bool buttonPressed = (digitalRead(buttonPin) == LOW);
60   bool isDark = (analogRead(ldrPin) < darknessThreshold);
61
62   // Start
63   if (buttonPressed && !isDark && !buttonPreviouslyPressed && !isPlaying) {
64     isPlaying = true;
65     currentNote = 0;
66     noteStartTime = millis();
67
68     // Turn LEDs ON for first note
69     for (int i = 0; i < 4; i++) digitalWrite(ledPins[i], HIGH);
70     ledsOn = true;
71
72     tone(buzzerPin, melody[currentNote], durations[currentNote]);
73     myservo.write(SLOW_SPEED);
74   }
75
76   if (isPlaying) {
```

TEST_FINALCODE.ino

```
72   tone(buzzerPin, melody[currentNote], durations[currentNote]);
73   myservo.write(SLOW_SPEED);
74 }
75
76 if (isPlaying) {
77   unsigned long now = millis();
78   int duration = durations[currentNote];
79
80   // ⚡ TURN LEDs OFF EXACTLY AFTER NOTE DURATION
81   if (ledsOn && (now - noteStartTime >= duration)) {
82     for (int i = 0; i < 4; i++) digitalWrite(ledPins[i], LOW);
83     ledsOn = false;
84   }
85
86   // 🏠 MOVE TO NEXT NOTE AFTER PAUSE
87   if (now - noteStartTime >= duration * 1.3) {
88     currentNote++;
89
90     if (currentNote < totalNotes) {
91       noteStartTime = now;
92
93       // Next note → LEDs ON again
94       for (int i = 0; i < 4; i++) digitalWrite(ledPins[i], HIGH);
95       ledsOn = true;
96
97       tone(buzzerPin, melody[currentNote], durations[currentNote]);
98     }
99     else {
100       noTone(buzzerPin);
101       myservo.write(STOP_SPEED);
102
103       for (int i = 0; i < 4; i++) digitalWrite(ledPins[i], LOW);
104
105       isPlaying = false;
106     }
107   }
108 }
109
110 if (!isPlaying && (!buttonPressed || !isDark)) {
111   myservo.write(STOP_SPEED);
112   for (int i = 0; i < 4; i++) digitalWrite(ledPins[i], LOW);
```

Time Spent

As Team: 19hr

Individually:

Andrew - 8hr

Kevin - 6hr

Cole - 5hr